

Ryan Liu

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EDUCATION

Northwestern University

B.S. Computer Science and Mathematics | GPA: 3.8/4.00

Expected Graduation: June 2027

Evanston, IL

Relevant Coursework: Data Structures, Algorithms, Machine Learning, Operating Systems, Databases, Computation Theory

AWARDS

ICPC North America Championship Honorable Mention ACM-ICPC

2025

USACO Platinum USA Computing Olympiad

2025

American Invitational Mathematics Exam (Top 0.27% of 300,000) Mathematical Association of America

2022, 2023

EXPERIENCE

Databricks

Incoming Software Engineer Intern

2026

Bellevue, WA

Amazon

Software Development Engineer Intern

June 2025 – Sep 2025

Seattle, WA

- Developed an end-to-end data pipeline to measure and report call network quality (latency, packet loss, audio jitter), addressing 300,000+ daily missed calls in NA alone and deployed the system to 5 AWS regions worldwide
- Integrated with CloudWatch, SNS, and Lambda to enable real-time alerting across global customer support centers
- Defined audio-quality scoring metrics and alert thresholds, standardizing call-quality monitoring across multiple organizations

Northwestern University

Guest Lecturer, Teaching Assistant

September 2024 – Present

Chicago, IL

- Selected to independently prepare and present the Winter 2025 COMP.SCI 212 (Discrete Math) guest lecture; Lectured 107 students on results in number theory and their applications to encryption

Juxta Technologies (YC S25)

Founding Software Engineer

June 2024 – Feb 2025

Chicago, IL

- Led the development of analytics-driven logistics tools for a \$6M health-tech startup backed by \$600K from Y Combinator, Lynett Capital, Velocity Catalyst, and Authentic Ventures, improving hospital efficiency and reducing operational costs
- Developed algorithms to minimize idle time, optimize task assignment, and streamline workflows using Python, JavaScript, and Firebase, boosting patient transporter efficiency by 44% and reducing transport times by 7.6 minutes per trip

Northwestern AI and Security Lab

Undergraduate Researcher

January 2024 – January 2025

Chicago, IL

- Built infrastructure to support prediction of time-to-exploitation and severity of Cybersecurity Vulnerabilities and Exposures
- Built CVE data pipelines from diverse datasets, leveraging APIs, web scraping, and data integration techniques

International Center for Advanced Internet Research

Information Technology Intern

October 2023 – August 2024

Chicago, IL

- Deployed and set up Data Transfer Nodes with NVMe over RDMA, achieving 400GB/s throughput
- Implemented a JupyterHub-based controller to detect DDoS attacks in real time using Sketch-based Entropy Estimation

PROJECTS

Offline Judge | Personal Project | [github](https://github.com) | 15 hours

HTTP Requests, Webscraping, C++, Python

- Developed an offline judge for olympiad problems, with test case scraping and runtime/memory limit enforcement
- Designed an system using g++, subprocess, and ulimit, ensuring accurate performance benchmarking

Customer Churn Prediction | Personal project | [github](https://github.com) | 15 hours

Python, Machine Learning, Data Processing

- Developed and deployed ML models, achieving 96% accuracy and 97% F1 score for churn prediction
- Conducted data analysis to identify key churn indicators like income levels and card preferences for actionable insights

SKILLS

Programming Languages: Python, C++, JavaScript, TypeScript, Java, SQL, MATLAB, R

Softwares and Tools: Git, Bash, React, React Native, Next.js, Node.js, HTML, CSS, NumPy, scikit-learn, OpenCV, REST APIs, GraphQL, Google Cloud Platform, AWS, Jupyter, PostgreSQL, MongoDB, Docker, CI/CD pipelines